

ALI JANLOO

AI Researcher & Machine Learning Engineer

📍 Tehran, Iran

📅 October 2000

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SUMMARY

Computer Science graduate with a specialized focus on Large Language Models, Agentic Workflows, and MLOps. I possess a track record of engineering production-grade AI applications (spanning RAG, speech, NLP, and computer vision systems). I combine theoretical knowledge from academic research with practical software engineering discipline to deploy scalable AI solutions, including topic modeling, hierarchical clustering, and insight generation for customer-history and chatbot analytics.

SKILLS

Languages & Core: Python, Cython, Bash, Linux, OOP, Clean Architecture

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, Scikit-Learn, spaCy, OpenCV, NumPy, Pandas

LLM Frameworks: LangChain, LangGraph, LlamaIndex, Ollama, MCP

Data Engineering: Apache Spark (PySpark), Kafka, RabbitMQ, Celery, Elasticsearch, MongoDB, PostgreSQL, Redis, Vector DBs (FAISS, Milvus)

MLOps & Cloud: FastAPI, Docker, AWS (ECS, SageMaker, Bedrock), GCP Vertex AI, Azure ML, GitHub Actions, MLflow, TensorRT, ONNX

EXPERIENCE

Mar 2026 – Present

AI Engineer

Smilino

- Built a topic modeling service for CRM-style file histories and active chatbot conversations on the Hamhoush platform.
- Developed hierarchical topic clustering and insight extraction pipelines to produce recommendations aligned with different managers' needs.
- Orchestrated batch and asynchronous processing with **Elasticsearch**, **MongoDB**, **RabbitMQ**, and **Celery** to support scalable analytics workflows.

Feb 2024 – Feb 2026

AI Engineer

ParsTech AI

- **RAG Optimization:** Diagnosed and resolved low retrieval accuracy in unstructured data pipelines by implementing a **hierarchical context-aware chunking** strategy. This ensured chunks retained parent-document context, significantly reducing LLM hallucinations (Improved the RAGAS Faithfulness by 20%).
- **Agentic Frameworks:** Architected 10+ **LangGraph** agents for the Parschat platform, including a semantic-powered Sales Assistant that leverages product attributes and user intent analysis for intelligent product filtering and recommendations.
- **Microservices:** Engineered a high-throughput FastAPI layer using Asyncio/Redis, reducing latency by **40%** and ensuring thread-safe handling of concurrent agent requests.
- **Infrastructure:** Deployed containerized services on AWS ECS/RDS with auto-scaling and established CI/CD pipelines via GitHub Actions.

Sep 2024 – Jan 2025

Teaching Assistant, Artificial Intelligence

Ferdowsi University of Mashhad

- Mentored undergraduate students in the "Fundamentals of Artificial Intelligence" course, focusing on search algorithms and optimization.
- Designed curriculum projects covering CSP, Evolutionary Algorithms, and Adversarial Search (Min-Max), bridging theory with Python implementation.

Mar 2023 – Jan 2024	AI Engineer	Vera110
	- Built an educational app for English language learners, developing Deep Learning models for ASR, Pronunciation Assessment, and Grammatical Error Correction. - Fine-tuned GPT-based language models for domain-specific tasks using LoRA (Low-Rank Adaptation) to improve educational content generation. - Optimized model inference speed by 1.5x using Knowledge Distillation and Post-Training Quantization (PTQ). - Built automated NLP workflows using spaCy to replace manual data entry, increasing data processing throughput by 60%.	
Feb 2021 – Mar 2023	AI & Software Engineer	Freelance
	- Engineered end-to-end AI solutions for SMEs, including custom chatbots and document analysis tools. - Built a fraud-scoring pipeline (Python/TensorFlow/Kafka) that reduced false positives by 30% while ensuring GDPR compliance. - Developed real-time ETL pipelines processing e-commerce logs via PySpark and RabbitMQ for instant analytics.	

PROJECTS

Retail Shelf Monitoring System

github.com/Alijanloo/Retail-Shelf-Monitoring

- Built a real-time system to detect Out-of-Stock (OOS) items using existing CCTV feeds.
- Leveraged **YOLOv12-M** for detection and **MobileNetV3 + FAISS** for high-speed SKU embedding search.
- Implemented frame alignment (ORB/SIFT + RANSAC) and SORT tracking to handle camera movement and occlusion.
- Achieved real-time inference on edge devices via **TensorRT/OpenVINO** optimization.

MultiModal Agentic RAG Pipeline

github.com/Alijanloo/MultiModalRag

- Architected a Multi-Modal RAG pipeline to index complex PDFs containing text, tables, and images.
- Integrated **Docling** and **TableFormer** for structural extraction and **Gemini-1.5-Flash** for image captioning.
- Utilized **LangGraph** for query reformulation and agentic reasoning to improve retrieval accuracy on ambiguous user queries.

Pdf2Table Library

github.com/Alijanloo/Pdf2Table

- Developed a Python library for structured table extraction from PDFs using Microsoft's Table Transformer models.
- The pipeline detects, parses, and reconstructs table grids, supporting both digital and scanned documents via TrOCR-based OCR.
- Outputs clean JSON representations of table structures, enabling seamless downstream use in NLP and data analysis workflows.

Other Public Projects

github.com/Alijanloo

Additional open-source projects and research code are available on my GitHub page.

PUBLICATIONS

[DOI](#)

A Novel Machine-Learning Based Method for Resolving Secondary Structure Topology in Medium-Resolution Cryo-EM
International Journal of Molecular Sciences
Co-author, 2026.

EDUCATION

2020 – 2025

B.Sc. in Computer Science
GPA: 3.31/4

Ferdowsi University of Mashhad

CERTIFICATIONS

[link](#)

Machine Learning Specialization

Stanford University (Andrew Ng)

Supervised/Unsupervised ML, neural networks, and ML pipelines.

[link](#)

Generative AI Engineering with LLMs

IBM

Focused on fine-tuning embedding and language models and building generative AI applications.

[link](#)

Digital Signal Processing: Basic Concepts and Algorithms

EPFL

Fundamentals of digital signal processing and analysis.

2024

IELTS

British Council

Overall Band: 6.5.

